

Zepto E-Commerce Data Analysis

PostgreSQL | pgAdmin

Dataset Size

- **Total Categories:** 15
- **Total Rows:** 3,777
- **Dataset Size:** 3,777 records across 15 categories.

About the Dataset

The dataset provided for this project is based on Zepto, an e-commerce platform. The data is divided into 15 different product categories, with each category maintained in a separate table.

Product Categories

1. Fruits & Vegetables 2. Cooking Essentials 3. Munchies 4. Dairy, Bread & Batter 5. Beverages 6. Packaged Food 7. Ice Cream & Desserts 8. Chocolates & Candies 9. Meats, Fish & Eggs 10. Biscuits 11. Personal Care 12. Paan Corner 13. Home & Cleaning 14. Health & Hygiene 15. Curated for You

Each category table contains the same column names:

1. Product_Name 2. MRP 3. Discount_Percent 4. Available_Quantity 5. Discount_Selling_Price
6. Weighted_In_GMS 7. Out_Of_Stock 8. Quantity

These columns provide information about the products, pricing, discounts, availability, weight, stock status, and quantity, making the dataset suitable for further analysis.

Project Objective

The main objective of this project is to analyze Zepto’s dataset (3,777 records across 15 categories) to uncover key insights into pricing strategies, stock availability, and category performance, with the goal of delivering actionable business recommendations..

Data Cleaning and Processing

The following steps were performed to organize, combine, and clean the product dataset:

1. Category Table Creation

A separate "categories" table was created with two columns: "category_id" and "category_name". All 15 product categories were added to this table.

2. Table Creation and Data Import

Separate tables were created in PostgreSQL for all 15 categories. The data was then imported into the respective PostgreSQL tables using pgAdmin.

Data Import Steps (pgAdmin):

- Left panel → Servers → Select the database
- Expand Schemas → Tables
- Right-click the target table → Import/Export Data
- Select Import and choose the CSV file
- Set encoding to UTF-8 and delimiter to comma (,)
- Keep Header ON and Freeze OFF
- Run the import

3. Master Product Table

After importing the data into the individual category tables, a master "product" table was created. Data from all 15 category tables was combined and stored in this table for further analysis.

4. Data Cleaning

The "product" table was checked for:

- Null or missing values
- Duplicate records
- Invalid values

A backup of the original product data was also created as "product_backup" before performing the cleaning operations.

5. Price Conversion

The original price values were provided in paise. These values were converted into rupees to make the data easier to understand and use for analysis.

6. Final Validation

The cleaned "product" table was checked again for null values and data consistency to ensure that the dataset was ready for further analysis.

Key Insights

1. Top 10 most expensive Products and with their available quantity

Product Name	MRP(₹)	Available Quantity
1.Borges Extra Light Olive Oil Bottle	2600	6
2.Borges Extra Light Olive Oil Bottle	2600	6
3.Pampers Pants - Large	1549	3
4.Pampers Pants - Large	1549	3
5.Praakritik Natural Desi Gir Cow A2 Ghee	1450	6
6.Praakritik Natural Desi Gir Cow A2 Ghee	1450	6
7.Dhara Kachi Ghani Mustard Oil Jar	1250	6
8.Dhara Kachi Ghani Mustard Oil Jar	1250	6
9.Saffola Gold (Jar)	1240	6
10.Saffola Gold (Jar)	1240	6

2. Top 5 most cheapest product with their availability quantity

Product Name	MRP(₹)	Available Quantity
1.Ching's Secret Paneer Chilli Masala	10	2
2.Ching's Secret Chicken 65 Masala - Chicken	10	5
3.Ching's Veg Manchurian Masala	10	2
4.Ching's Chow Mein Hakka Noodles Masala.	10	0
5.Ching's Secret Chicken 65 Masala - Chicken.	10	5

3. Category-wise discount analysis: Ranking all 15 categories from highest to lowest average discount.

Category Name	Average_discount
1. Curated For You	15.89
2. Fruits & Vegetables	15.46
3. Meats, Fishes & Eggs	11.03
4. Icecream & Desserts	8.32
5. Chocolates & Candies	8.32
6. Packaged Food	8.32
7. Biscuits.	8.24
8. Health & Hygience	8.05
9. Cooking Essentials	7.16
10. Beverages.	7.16
11. Dairy, Bread & Batter	7.16
12. Munchies	7.16
13. Personal Care	6.25
14. Paan Corner	6.25
15. Home & Cleaning	5.70

4. Total MRP vs total discounted price per category (overall discount impact)

Category Name	total product price (₹)	total discount	after discount price(₹)	total savings (₹)
1. Cooking Essentials	80459	3682	72385	8074
2. Munchies	80459	3682	72385	8074
3. Personal Care	71249	2150	65270	5979
4. Paan Corner	71249	2150	65270	5979
5. Packaged Food	60422	3230	54738	5684
6. Icecream & Desserts	60422	3230	54738	5684
7. Chocolates & Candies	60422	3230	54738	5684
8. Home & Cleaning	29518	1101	27441	2077
9. Health & Hygience	15440	781	14148	1292
10. Beverages	15313	923	13381	1932
11. Dairy, Bread & Batter	15313	923	13381	1932
12. Meats, Fishes & Eggs	11799	695	10338	1461
13. Biscuits	8745	1212	7761	984
14. Fruits & Vegetables	4372	1438	3705	667
15. Curated For You	3611	731	3041	570

5. Estimated Revenue Generated by each category

Category Name	Total sales(₹)
1.Cooking Essentials	337369
2.Munchies	337369
3.Personal Care	270849
4.Paan Corner	270849
5.Packaged Food	224385
6.Icecream & Desserts	224385
7.Chocolates & Candies	224385
8.Home & Cleaning	122661
9.Health & Hygiene	64180
10.Beverages	55051
11.Dairy, Bread & Batter	55051
12.Biscuits	25019
13.Meats, Fishes & Eggs	20693
14.Curated For You	16648
15.Fruits & Vegetables	10846

6. High MRP products that are out of stock (lost sales opportunities

There are total **454** rows but in this we only represent 7 top rows

Category Name	MRP(₹)
1.Patanjali Cow's Ghee	565
2.Patanjali Cow's Ghee	565
3.MamyPoko Pants Standard Diapers, Extra Large (12 - 17 kg)	399
4.MamyPoko Pants Standard Diapers, Extra Large (12 - 17 kg)	399
5.Aashirvaad Atta With Mutigrains	315
6.Aashirvaad Atta With Mutigrains	315
7.Everest Kashmiri Lal Chilli Powder	310

7. Top 10 discounted Product

Category Name	MRP(₹)	discount (%)	dis_selling_price(₹)
1.Dukes Waffy Chocolate Wafers	45	51	22
2.Dukes Waffy Orange Wafers	45	51	22
3.Dukes Waffy Strawberry Wafers	45	51	22
4.Chef's Basket Durum Wheat Penne Pasta	160	50	80
5.Chef's Basket Durum Wheat Fusilli Pasta	160	50	80
6.Ceres Foods Fish Mustard Instant Liquid Masala	220	50	110
7.Ceres Foods Nalli Nihari Instant Liquid Masala	220	50	110
8.Ceres Foods Laal Maas Instant Liquid Masala	220	50	110
9.Chef's Basket Durum Wheat Elbow Pasta	160	50	80
10.Chef's Basket Durum Wheat Elbow Pasta	160	50	80

8. Product with no discount at all

There are total 1179 rows but in this we only represent top 10

Product Name	MRP(₹)	total discount(%)
1. Johnson's Baby Oil	200	0
2. Maggi 2 Minute Masala Noodles	12	0
3. Tata Salt	24	0
4. Arden Eggs White	97	0
5. Saffola Gold Oil (Pouch)	215	0
6. Fortune Sunlite Refined Sunflower (Jar)	925	0
7. Saffola Active Oil	210	0
8. Fortune Soyabean Oil	1005	0
9. Popular Essentials Sunflower Seeds	25	0
10. Tata Sampann High Protien Kabuli Chana	94	0

9. Total Number of products in each category

Category Name	Total product
1. Munchies	514
2. Cooking Essentials	514
3. Packaged Food	388
4. Icecream & Desserts	388
5. Chocolates & Candies	388
6. Paan Corner	344
7. Personal Care	344
8. Home & Cleaning	193
9. Biscuits	147
10. Beverages	129
11. Dairy, Bread & Batter	129
12. Health & Hygience	97
13. Fruits & Vegetables	93
14. Meats, Fishes & Eggs	63
15. Curated For You	46

10. Top 3 products giving customers the highest savings

Product Name	Total savings
1. Borges Extra Light Olive Oil Bottle	1201
2. Pampers Pants - Large	700
3. Kelloggs Chocos - Moons & Stars	264

Recommendations

1 Revenue contribution % of each category to total business revenue

Category Name	Total Sales.	Business Contribution
1.Cooking Essentials	337369	14.93
2.Munchies.	337369	14.93
3.Personal Care	270849	11.99
4.Paan Corner	270849	11.99
5.Packaged Food	224385	9.93
6.Icecream & Desserts	224385	9.93
7.Chocolates & Candies	224385	9.93
8.Home & Cleaning	122661	5.43
9.Health & Hygience	64180	2.84
10.Beverages	55051	2.44
11.Dairy, Bread & Batter	55051	2.44
12.Biscuits	25019	1.11
13.Meats, Fishes & Eggs	20693	0.92
14.Curated For You	16648	0.74
15.Fruits & Vegetables	10846	0.48

2. Stock health per category (which portion of a category catalog is out of stock)

Category Name	Total Products	Out of stocks
1.Munchies	514	64
2.Cooking Essentials	514	64
3.Packaged Food	388	45
4.Icecream & Desserts.	388	45
5.Chocolates & Candies	388	45
6.Biscuits	147	42
7.Beverages	129	28
8.Dairy, Bread & Batter	129	28
9.Personal Care	344	21
10.Paan Corner	344	21
11.Home & Cleaning	193	19
12.Health & Hygience	97	13
13.Meats, Fishes & Eggs	63	12
14.Fruits & Vegetables	93	6
15.Curated For You	46	1

3. Total number of products having low stock , high demand risk products

There are total **1665** products

4. Price range/ bucket distribution of products

Price bucket	product count
1.Under 100	1877
2.100-300	1600
3.301-600	242
4.601-1000	44
5.Above 1000	14

5. Correlation check - does higher discount means higher out of stock rate

Discount_bucket	total products	out of stock	out of stock rate
1.No Discount	1179	176	14.93
2.Above 40%	69	9	13.04
3.1-20%	2350	261	11.11
4.21-40%	179	8	4.47

6. How many products with MRP and selling price almost equal (negligible discount , <5%)

There are total **443** products

7. Overall business summary in one query (single row dashboard snapshot)

- 1. **Total Products** : 3777
- 2. **Avg discount percent** : 7.72
- 3. **total customer savings** : 56073
- 4. **total out of stock** : 454
- 5. **out of stock rate percent** : 12.02



zepto

RECOMMENDATIONS FOR ZEPTO

Smart Strategies for Better Sales,
Inventory & Customer Satisfaction



1 CLEARANCE STRATEGY FOR SLOW-MOVING PRODUCTS

For products with low sales that are approaching their expiry date, offer Buy One Get One Free (BOGO) deals or attractive discounts to clear inventory and minimize product wastage.



2 MINIMAL DISCOUNTS ON ESSENTIAL PRODUCTS

For essential and high-demand products that customers regularly need, maintain minimal discounts since their demand is relatively stable. This helps protect profit margins while maintaining sales.



3 BUNDLE HIGH-INVENTORY PRODUCTS

Products with excess inventory can be combined with frequently purchased products into attractive bundles. This helps increase sales while reducing excess stock.



4 IMPROVE AVAILABILITY OF HIGH-DEMAND PRODUCTS

Products that frequently go out of stock should be monitored closely and given higher inventory priority. Maintaining sufficient stock can prevent lost sales and improve customer satisfaction.



5 TARGETED DISCOUNTS INSTEAD OF BLANKET DISCOUNTS

Discounts should be focused on low-performing or slow-moving products rather than applying discounts across the entire catalog. This increases sales where needed while protecting margins on products that already have strong demand.